`backend/core/hardware.py` Documentation (Roman Urdu)

Yeh ek utility (helpful tool) script file hai jiska asaan maqsad system hardware (GPU aur RAM) ko detect karna aur check karna hai. Application fine-tuning aur large inference process kartay huway hardware capacity par rely karti hai taake computer freeze ya crash na ho.

File Ka Maqsad (Purpose)

Is me wo logic likhi gayi hai jo Windows aur Linux systems par hardware resources ko measure karti hai. Application models (Llama, Mistral) ko 4-bit (choti space me load krnay) ya CPU fallback karne ka failsafe idher dhoondti ha.

Code Blocks aur Functions ki Tafseel

1. `def get_gpu_memory_gb() -> float | None`

- **Kya karta hai?**: Yeh function system mein mojud Graphic Card (Dedicated GPU / VRAM) ki total memory measure karta hai aur usey Gigabytes (GB) mein return krta hai. Agar server pe GPU available na ho (sirf cpu ho) to yeh error throw karne k bjaay simply `None` return kardeta hai.

  Tareeqa e kaar (Logic Flow): Yeh function memory maloom karne ke lie 3 alag tareeqay b-taur fallbacks istemal karta hai kiyun k hardware detection mukhtalif computer OS par fail ho skti ha:

  1. **GPUtil**: Pehley package library try krta ha (NVIDIA cards k liy easily kaam kr jata hai).
  2. **`nvidia-smi` Command line**: Agar pehla tareeqa na challay toh direct shell / console pr NVIDIA ka SMI tool chala k vram parhne ki koshish karta hai (`subprocess.run` k zariye).
  3. **PyTorch fallback (`torch.cuda`)**: Agar upr wale dono options kaamm na ayen toh AI system ki PyTorch library k default cuda tools istamal kr k VRAM access hasil krta hai.

2. `def get_cpu_ram_gb() -> float`

- **Kya karta hai?**: Yeh system ka Standard RAM (Random Access Memory) calculate karta hai Gigabytes mein.

- **Tareeqa e kaar (Logic Flow)**: Ye simply `psutil` namoolam utility library use kar k `psutil.virtual_memory().total` command call krta ha, bit / byte values ko divide karta hai `1024 ** 3` par aur user memory ka saaf figure (jaise *16.0 GB* ya *32.0 GB*) return karta hai.